

ARTIFICIAL INTELLIGENCE IN FINTECH / WEEK 03

From Prices to Technical Indicators

Calculate each indicator from real prices, then see how traders commonly use it. Distinguish a possible trading condition from a proven strategy, and check when its inputs become available.

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Read each question, calculate from the same real prices, then explain the output. Setup is on page 2. Questions are part of the lesson, not separate submissions.

Main data: AAPL, January 2-June 28, 2024. Issues 16-17 add earlier AAPL history. Prices are USD/share; returns exclude dividends and costs. Short windows, MACD 12/26/9 and Wilder 14 are teaching settings, not recommendations.

This week does not rank trading performance or choose portfolio weights. Shown dates are selected teaching examples, not estimates of how frequently a pattern occurs.

DATA PREPARATION

Prepare the common AAPL observations

Reuse Week 2's field and date conventions. Keep Close separate from Adj Close and run the examples in order. The short checks below stop calculation rather than silently changing prices.

Before running: Will July 1 appear when it is the exclusive end date?

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import yfinance as yf

aapl = yf.Ticker("AAPL").history(
    start="2024-01-02", end="2024-07-01",
    interval="1d", auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
ohlcv = aapl[["Open", "High", "Low", "Close"]]
assert aapl.index.is_unique and aapl.index.is_monotonic_increasing
assert np.isfinite(ohlcv).all().all() and (ohlcv > 0).all().all()
assert (aapl.High >= ohlcv[["Open", "Close"]].max(axis=1)).all()
assert (aapl.Low <= ohlcv[["Open", "Close"]].min(axis=1)).all()
assert np.isfinite(aapl.Volume).all() and (aapl.Volume >= 0).all()
print("Rows:", len(close), "| Dates:", close.index[0], close.index[-1])
result = pd.DataFrame({"Date": close.index[:3],
    "Close (USD/share)": close.iloc[:3].map("{:.2f}".format)})
result
```

Rows: 124 | Dates: 2024-01-02 2024-06-28

Date	Close (USD/share)
2024-01-02	185.64
2024-01-03	184.25
2024-01-04	181.91

There are 124 returned rows. January 2-4 closes are USD 185.64, 184.25 and 181.91/share. Reuse the response instead of changing the sample for each indicator.

These checks do not verify every expected exchange date or historical receipt time. The request returns today's historical version, not a log of data arriving in 2024. Missing lookback history is not zero.

Install numpy, pandas, matplotlib and yfinance if needed. All displayed code runs in one fresh Python notebook; tables are each block's final expression. Outputs below were actually executed. Formulas use rounded inputs; code retains precision.

Sources: [1], [17].

ISSUE 01

What does a moving average measure?

A simple moving average (SMA) shows the average of recent closing prices. Use three prices to see exactly what enters the average.

C_t is Close in USD/share. k counts prices; j counts rows backward from t . Use $k = 3$.

$$SMA_t^{(k)} = \frac{1}{k} \sum_{j=0}^{k-1} C_{t-j}$$

$$SMA_{Jan4}^{(3)} = \frac{185.64 + 184.25 + 181.91}{3} \approx 183.93 \text{ USD/share}$$

Before running: Which is the first date with three prices? Will its Close be below the average?

```
sma3 = close.rolling(3, min_periods=3).mean()
result = pd.DataFrame({"Date": close.index[:5],
    "Close (USD)": close.iloc[:5].map("{:.2f}".format),
    "SMA3 (USD)": sma3.iloc[:5].map(
        lambda x: "Missing" if pd.isna(x) else f"{x:.4f}")})
result
```

Date	Close (USD)	SMA3 (USD)
2024-01-02	185.64	Missing
2024-01-03	184.25	Missing
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833

AAPL prices and SMA are USD/share. January 4 uses January 2, 3 and 4: the mean is 183.9333, above the Close of 181.91. The first two means are missing because the window is incomplete.

Common use: a price crossing above its SMA can flag a possible entry. January 5 Close 181.18 is below SMA3 182.4467; January 8 Close 185.56 is above SMA3 182.8833. This flags January 8 after its Close arrives, not a guaranteed gain or fill at that Close. [10]

Discuss

1. Calculate January 5's SMA using the table. Identify the price removed and the price added.
2. Why would filling the first two missing SMA values with zero misrepresent the data? State how many prices each row actually has.

Sources: [3], [10]. Full titles and links are on the references page.

ISSUE 02

Can the Close fall while its moving average rises?

The daily Close change compares today with yesterday. The SMA change compares the price entering its window with the price leaving it.

C is Close in USD/share. For a three-price SMA, subtract the two averages: the two shared prices cancel.

$$\frac{C_t + C_{t-1} + C_{t-2}}{3} - \frac{C_{t-1} + C_{t-2} + C_{t-3}}{3} = \frac{C_t - C_{t-3}}{3}$$

January 9 adds Close 185.14 and removes January 4 Close 181.91 USD/share.

$$SMA_{Jan9}^{(3)} - SMA_{Jan8}^{(3)} = \frac{185.14 - 181.91}{3} \approx 1.0767 \text{ USD/share}$$

Before running: Can the Close column fall while the SMA3 column rises on January 9?

```
dates = close.loc["2024-01-02":"2024-01-10"].index
result = pd.DataFrame({"Date": dates,
    "Close (USD)": [f"{close[d]:.2f}" for d in dates],
    "SMA3 (USD)": ["Missing" if pd.isna(sma3[d])
        else f"{sma3[d]:.4f}" for d in dates]})
result
```

Date	Close (USD)	SMA3 (USD)
2024-01-02	185.64	Missing
2024-01-03	184.25	Missing
2024-01-04	181.91	183.9333
2024-01-05	181.18	182.4467
2024-01-08	185.56	182.8833
2024-01-09	185.14	183.9600
2024-01-10	186.19	185.6300

Close falls USD 0.42/share, but SMA3 rises from USD 182.8833 to USD 183.9600/share. The new price exceeds the discarded price.

Common use: a rising SMA can restrict a strategy to an upward trend. A rule requiring SMA3 to rise still passes on January 9 (+1.0767 USD/share), despite the daily Close falling 0.42. The rule describes its chosen trend, not every daily return. [10]

Discuss

1. Write the dates in the January 8 and 9 averages. Which two prices cancel?
2. Use the January 8-9 numbers to correct the claim that a rising SMA means the stock rose that day.

Sources: [3], [10]. Full titles and links are on the references page.

ISSUE 03

How does an exponential average update?

An exponential moving average (EMA) combines today's Close with yesterday's EMA. It gives more weight to recent prices.

C_t and E_t are today's Close and EMA, in USD/share. Alpha is the weight on today's Close. Use span 3, alpha 0.5 and the first Close as the initial EMA.

$$E_t = \alpha C_t + (1 - \alpha)E_{t-1}, \quad \alpha = \frac{2}{\text{span} + 1}$$

$$\alpha = \frac{2}{3+1} = 0.5; \quad E_{\text{Jan3}} = 0.5(184.25) + 0.5(185.64) = 184.945$$

$$E_{\text{Jan4}} = 0.5(181.91) + 0.5(184.945) = 183.4275 \text{ USD/share}$$

Before running: Will January 4 EMA be halfway between its Close and January 3 EMA?

```
ema3 = close.ewm(span=3, adjust=False,
    min_periods=1, ignore_na=False).mean()
dates = close.index[:7]
result = pd.DataFrame({"Date": dates,
    "Close (USD)": close.loc[dates].map("{:.2f}".format),
    "Previous EMA": ema3.shift(1).loc[dates].map(
        lambda x: "Initial value" if pd.isna(x) else f"{x:.4f}"),
    "EMA3 (USD)": ema3.loc[dates].map("{:.4f}".format)})
result
```

Date	Close (USD)	Previous EMA	EMA3 (USD)
2024-01-02	185.64	Initial value	185.6400
2024-01-03	184.25	185.6400	184.9450
2024-01-04	181.91	184.9450	183.4275
2024-01-05	181.18	183.4275	182.3037
2024-01-08	185.56	182.3037	183.9319
2024-01-09	185.14	183.9319	184.5359
2024-01-10	186.19	184.5359	185.3630

January 4: halfway between 181.9100 and 184.9450 is 183.4275 USD/share. January 5 then uses 183.4275 as its previous EMA. All EMA values, including initialization, are visible.

adjust=False selects this recursion. Showing the initial rows does not make the EMA fully warmed up; a different starting history can give different values.

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Which past prices still affect the EMA?

Expand January 4's update once. The earlier EMA already contains January 2 and 3 prices, so the newest result retains both.

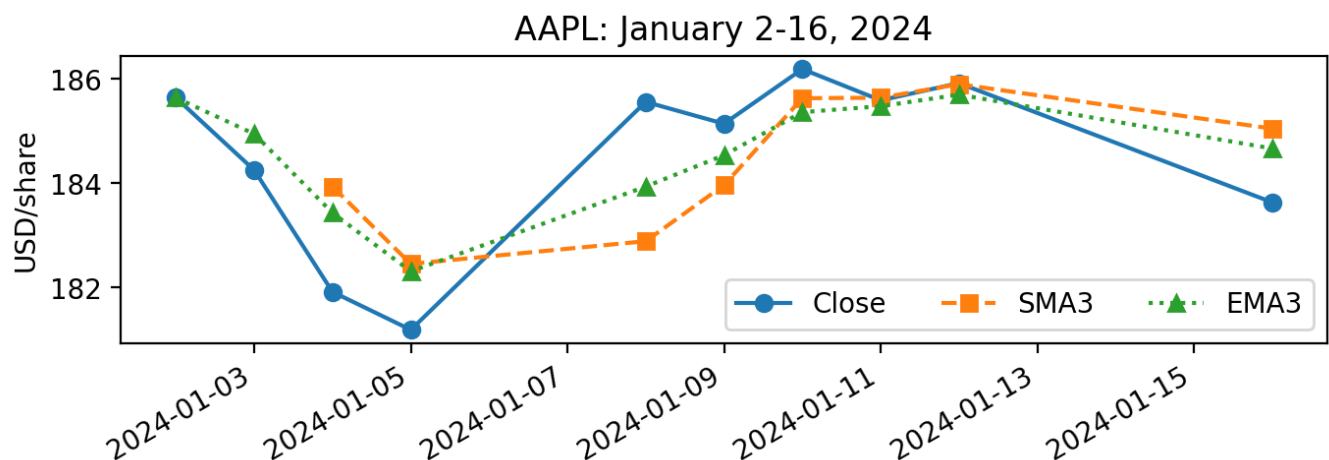
$$E_{\text{Jan4}} = 0.5C_{\text{Jan4}} + 0.25C_{\text{Jan3}} + 0.25C_{\text{Jan2}}$$

$$E_{\text{Jan4}} = 0.5(181.91) + 0.25(184.25) + 0.25(185.64) = 183.4275$$

The weights sum to 1. On January 5, each old weight is halved again; January 2 still has weight 0.125. SMA3 drops January 2 completely.

Before running: On January 8, Close jumps to 185.56. Will EMA3 or SMA3 be closer to that Close?

```
fig, ax = plt.subplots(figsize=(7, 2.6))
dates = pd.to_datetime(close.index[:10])
ax.plot(dates, close.iloc[:10], "o-", label="Close")
ax.plot(dates, sma3.iloc[:10], "s--", label="SMA3")
ax.plot(dates, ema3.iloc[:10], "^:", label="EMA3")
ax.set(title="AAPL: January 2-16, 2024", ylabel="USD/share")
ax.legend(ncol=3)
fig.autofmt_xdate()
plt.tight_layout()
```



January 8: Close 185.56, EMA3 183.9319, SMA3 182.8833 USD/share. The EMA gives the newest price weight 0.5 rather than 1/3, so it is closer here. That is a response to observed prices, not a forecast.

Common use: traders use a rising EMA to identify an upward trend. An illustrative rule considers buying only when EMA3 rises: January 8 passes because 183.9319 exceeds January 5 EMA 182.3037. Faster response can also produce more frequent changes of direction; it is not automatically a more profitable rule. [11]

Discuss

1. Expand January 4's recursion to find the weights on January 2, 3 and 4. Do they sum to one?
2. Why must a classmate state the seed and adjust setting before comparing their EMA with yours? Does span=3 mean exactly three prices are used?

Sources: [4], [11]. Full titles and links are on the references page.

ISSUE 04

What do MACD and its histogram compare?

Moving Average Convergence Divergence (MACD) measures the gap between a shorter and a longer price average. It helps describe how those averages move together or apart.

$E_t^{(12)}$ and $E_t^{(26)}$ are Close EMAs with spans 12 and 26. Both start at January 2 Close and use `adjust=False`. Their difference, MACD, is in USD/share.

$$\text{MACD}_t = E_t^{(12)} - E_t^{(26)}$$

$$\text{MACD}_{\text{Apr25}} = 168.86346 - 170.47263 \approx -1.6092$$

Before running: Will April 25 MACD be positive when EMA12 is below EMA26?

```
ema12 = close.ewm(span=12, adjust=False).mean()
ema26 = close.ewm(span=26, adjust=False).mean()
macd = ema12 - ema26
signal_line = macd.ewm(span=9, adjust=False).mean()
histogram = macd - signal_line
dates = close.loc["2024-04-23":"2024-04-26"].index
result = pd.DataFrame({"Date": dates,
    "EMA12": [f"{ema12[d]:.4f}" for d in dates],
    "EMA26": [f"{ema26[d]:.4f}" for d in dates],
    "MACD": [f"{macd[d]:.4f}" for d in dates]})
result
```

Date	EMA12	EMA26	MACD
2024-04-23	168.6144	170.6392	-2.0248
2024-04-24	168.6768	170.5192	-1.8424
2024-04-25	168.8635	170.4726	-1.6092
2024-04-26	168.9306	170.3858	-1.4552

All table values are USD/share. On April 25, EMA12 is 1.6092 below EMA26. Negative MACD means the shorter average is lower; it does not mean that day's return is negative.

The signal line is a 9-span EMA of MACD, not an EMA of Close. The code calculates it here for the next page. Its initial value is the first MACD, zero; earlier starting history may change the results.

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Why show a signal line and histogram?

S_t denotes the signal line in USD/share. Its smoothing weight is $2/(9+1)=0.2$. The histogram displays the signed gap between MACD and this smoother line.

$$S_t = 0.2 \text{MACD}_t + 0.8S_{t-1}, \quad \text{Histogram}_t = \text{MACD}_t - S_t$$

$$S_{\text{Apr25}} \approx 0.2(-1.60917) + 0.8(-1.81626) \approx -1.77484$$

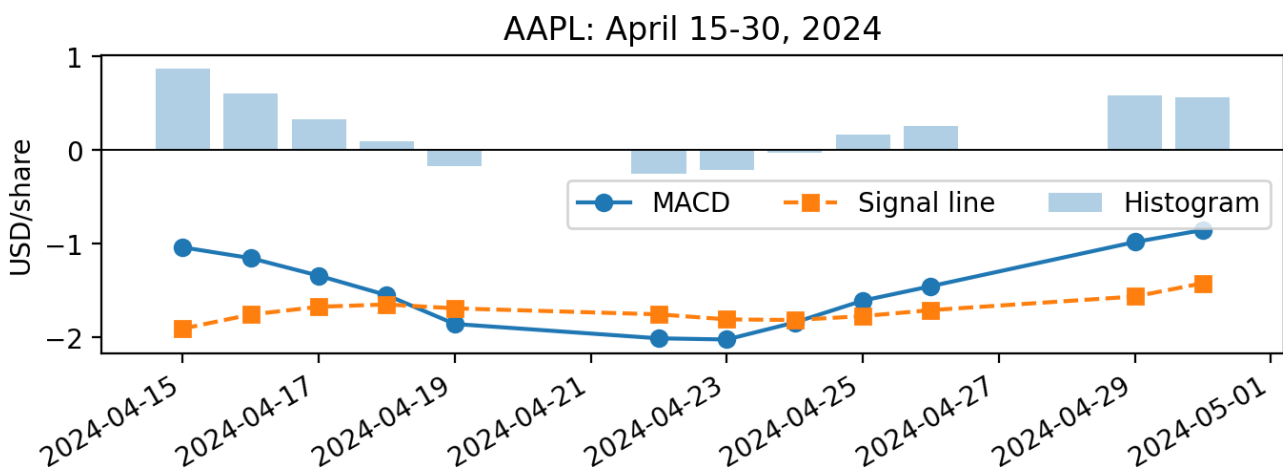
$$\text{Histogram}_{\text{Apr25}} \approx -1.60917 - (-1.77484) \approx 0.1657$$

Before running: Will April 25 have a positive bar even though both lines are below zero?

```

window = slice("2024-04-15", "2024-04-30")
dates = pd.to_datetime(macd.loc[window].index)
fig, ax = plt.subplots(figsize=(7, 2.7))
ax.plot(dates, macd.loc[window], "o-", label="MACD")
ax.plot(dates, signal_line.loc[window], "s--", label="Signal line")
ax.bar(dates, histogram.loc[window], alpha=0.35, label="Histogram")
ax.axhline(0, color="black", linewidth=0.7)
ax.set(title="AAPL: April 15-30, 2024", ylabel="USD/share")
ax.legend(ncol=3)
fig.autofmt_xdate()
plt.tight_layout()

```



April 24 histogram is -0.0262; April 25 is +0.1657 USD/share. MACD crossed above S. The bars show this crossing directly; they add no new market observation.

Common use: this upward crossing can flag a possible buy entry. The example flags April 25 even though MACD remains negative. A rule requiring MACD above zero would not pass. These are different conditions; repeated crossings can create losing trades. [18]

Discuss

1. On April 25, explain negative MACD and positive histogram using the two different comparisons.
2. If the histogram equals zero, what must be true of MACD and S? Does either price EMA have to equal the other?

Sources: [4], [18], [23]. Full titles and links are on the references page.

ISSUE 05

Why do short and longer horizons give opposite signs?

On January 9, AAPL had recovered from January 4 but was still below January 2. Different starting prices therefore give different return signs.

C is Close in USD/share. h counts trading-observation intervals, not calendar days. m is the price return ending on the same date t.

$$m_t^{(h)} = \frac{C_t}{C_{t-h}} - 1$$

$$m_{\text{Jan9}}^{(3)} = \frac{185.14}{181.91} - 1 \approx 1.7756\%$$

$$m_{\text{Jan9}}^{(5)} = \frac{185.14}{185.64} - 1 \approx -0.2693\%$$

Before running: Will the two January 9 return rows have opposite signs despite the same ending Close?

```
momentum3 = close / close.shift(3) - 1
momentum5 = close / close.shift(5) - 1
starts = ["2024-01-04", "2024-01-02"]
end = "2024-01-09"
result = pd.DataFrame({"Start date": starts,
    "Intervals": [3, 5],
    "Start C": [f"{close[d]:.2f}" for d in starts],
    "Jan 9 C": [f"{close[end]:.2f}"] * 2,
    "Return": [f"{close[end]/close[d]-1:.4%}" for d in starts]})
result
```

Start date	Intervals	Start C	Jan 9 C	Return
2024-01-04	3	181.91	185.14	1.7756%
2024-01-02	5	185.64	185.14	-0.2693%

January 9 returns are 1.7756% and -0.2693%. Price has recovered above January 4 but remains below January 2. No contradictory measurement is involved.

Common use: require positive momentum before considering a purchase. January 9 passes with three intervals (+1.7756%) but fails with five (-0.2693%). Fix the horizon before applying the rule; changing it after seeing the answer changes the rule. [13]

Discuss

1. Use the three dated closes to explain how both January 9 results can be correct. Which period does each describe?
2. Would choosing whichever horizon has the higher return demonstrate predictive skill? State what must be fixed before testing later returns.

Sources: [5], [13]. Full titles and links are on the references page.

ISSUE 06

How dispersed were the recent returns?

Volatility describes dispersion in returns, not whether prices rose overall.

r_1, r_2, r_3 are decimal returns ending January 3, 4 and 5. $\bar{r}$ is their mean. Sample standard deviation uses $n-1=2$ for these $n=3$ returns; its unit is return, not dollars.

$$\bar{r} = \frac{r_1 + r_2 + r_3}{3}, \quad \hat{\sigma} = \sqrt{\frac{(r_1 - \bar{r})^2 + (r_2 - \bar{r})^2 + (r_3 - \bar{r})^2}{2}}$$

$$\bar{r} = \frac{-0.0074876 - 0.0127001 - 0.0040130}{3} \approx -0.0080669$$

$$\hat{\sigma} \approx \sqrt{\frac{(0.0005793)^2 + (-0.0046332)^2 + (0.0040539)^2}{2}} \approx 0.004372 = 0.4372\%$$

Squaring prevents positive and negative deviations from cancelling. The square root restores the return unit.

For a hypothetical USD 1,000 exposure, multiply each return by 1,000. Its standard deviation scales by 1,000 too; the last column expresses this in USD.

Before running: Will the January 5 rolling value reproduce the three-return hand calculation? Does its positive sign mean a positive price return?

```
daily_return = close.pct_change(fill_method=None)
vol3 = daily_return.rolling(3, min_periods=3).std(ddof=1)
dates = ["2024-01-05", "2024-01-08"]
result = pd.DataFrame({"Ending date": dates,
    "3-return volatility": [f"{vol3[d]:.4%}" for d in dates],
    "Std. dev. (USD)": [f"{1000 * vol3[d]:.2f}" for d in dates]})
result
```

Ending date	3-return volatility	Std. dev. (USD)
2024-01-05	0.4372%	4.37
2024-01-08	1.9278%	19.28

The January 5 sample standard deviation is 0.4372% per interval. Its sign is nonnegative even though all three daily returns were negative.

Common use: translate return variability into dollars when assessing exposure. With a fixed USD 1,000 starting value, 0.4372% corresponds to USD 4.37; January 8's 1.9278% corresponds to USD 19.28. This compares historical dispersion at the same dollar exposure, not maximum loss or tomorrow's expected move. [26]

Discuss

1. Why can three negative returns have positive standard deviation? Explain centering, squaring and the square root.
2. Why do three returns need four prices? What history changes in a 20-return window?

Sources: [3], [14], [26]. Full titles and links are on the references page.

ISSUE 07

Why include yesterday's Close?

One share can fall further from yesterday's Close than today's High-Low range. Use the same AAPL prices to see why.

Assume you already held one share at the January 3 Close. No purchase or sale is being simulated. All prices below are USD/share.

Date	Observation	Price (USD)
2024-01-03	Close	184.25
2024-01-04	High	183.09
2024-01-04	Low	180.88

From yesterday's Close to today's Low

$$184.25 - 180.88 = 3.37 \text{ USD/share}$$

At the Low, your one share is worth USD 3.37 less than at the previous Close.

From today's High to today's Low

$$183.09 - 180.88 = 2.21 \text{ USD/share}$$

This is a smaller distance because today's High, 183.09, is already below yesterday's Close, 184.25.

The point: High-Low leaves out the earlier Close.

The two answers use different starting prices. To measure a range that also includes the previous Close, we use True Range (TR). [21]

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TR includes the previous Close

TR measures the largest distance among today's High, today's Low and the previous Close. On January 4, the furthest-apart prices are 184.25 and 180.88, so TR is 3.37 USD/share.

H and L are today's High and Low; C is Close. t is January 4 and t-1 is January 3. All three inputs are USD/share.

Date	Previous C	H	L
2024-01-04	184.25	183.09	180.88

The standard formula checks all three pairs. Absolute values turn a difference into a nonnegative distance.

$$TR_t = \max\{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\}$$

$$TR_{\text{Jan4}} = \max\{183.09 - 180.88, |183.09 - 184.25|, |180.88 - 184.25|\}$$

$$TR_{\text{Jan4}} = \max\{2.21, 1.16, 3.37\} = 3.37 \text{ USD/share}$$

Before running: Will the January 4 TR output be 2.21 or 3.37 USD/share?

```
previous_close = close.shift(1)
distances = pd.concat([
    aapl["High"] - aapl["Low"],
    (aapl["High"] - previous_close).abs(),
    (aapl["Low"] - previous_close).abs()], axis=1)
# Jan 2 has no previous Close in this sample.
true_range = distances.iloc[1:].max(axis=1)
date = "2024-01-04"
result = pd.DataFrame({"Date": [date],
    "High-Low (USD)": [f"{distances.loc[date,0]:.2f}"],
    "TR (USD)": [f"{true_range[date]:.2f}"]})
result
```

Date	High-Low (USD)	TR (USD)
2024-01-04	2.21	3.37

The output is 3.37, matching the largest of the three distances. TR includes the previous Close; High-Low, 2.21, does not.

The point: TR measures movement size, not a buy or sell direction.

Its positive value does not mean you earned money. It is not a limit on tomorrow's loss. Sources: [19], [21].

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Use ATR to choose a stop distance

Common use: traders smooth daily TR values into Average True Range (ATR), then use a multiple of ATR to set a stop distance. This avoids using only one day's unusually large or small range. [21]

For this example, use the January 24 ATR14 value below. Issue 8 explains how it is calculated from daily TR. Prices and ATR are USD/share.

Date	Close	ATR14
2024-01-24	194.50	3.2571

Suppose you own the share and choose a sell stop 2 ATR below this Close. A sell stop triggers a market sell order if the stop price is reached; it is not a guaranteed sale price. [27]

C is the reference Close; k is the chosen multiplier, here 2. These settings are an illustration, not an optimized trading rule.

$$\text{Stop distance} = k \text{ ATR}_t, \quad \text{Stop price} = C_t - k \text{ ATR}_t$$

$$\text{Distance} = 2(3.2571) = 6.5142 \text{ USD/share}$$

$$\text{Stop price} = 194.50 - 6.5142 \approx 187.99 \text{ USD/share}$$

The point: ATR turns recent movement into a dollar distance.

Here the chosen stop is about USD 6.51 below the reference Close. A larger ATR would put the stop further away at the same multiplier. The January 24 inputs must arrive before this level can be set.

A gap below the stop can lead to a lower execution price. The calculation does not guarantee a USD 6.51 loss limit. [27]

Discuss

1. January 4 High-Low is 2.21, but TR is 3.37 USD/share. Which starting prices explain the difference? Use 183.09 and 184.25.

2. Keep Close 194.50 and ATR 3.2571 fixed. Calculate a 1-ATR stop instead. Would it trigger after a smaller or larger price fall than the 2-ATR stop?

Sources: [19], [21], [27]. Full titles and links are on the references page.

ISSUE 08

What does ATR tell us about price movement?

Average True Range (ATR) summarizes recent price-movement size, including gaps. It helps distinguish quieter periods from larger daily movements; it does not say which direction prices will move.

Use Wilder's 14-period method. The first ATR is the mean of the first 14 valid TR values. Both quantities are USD/share; i indexes the 14 dates below.

$$ATR_{\text{Jan23}} = \frac{1}{14} \sum_{i=1}^{14} TR_i$$

$$ATR_{\text{Jan23}} \approx 46.9100/14 \approx 3.3507 \text{ USD/share}$$

Before running: Which date completes 14 valid TR values? Can ATR14 be initialized earlier?

```
seed_tr = true_range.iloc[:14]
result = pd.DataFrame({"Date": seed_tr.index,
                      "TR (USD/share)": seed_tr.map("{:.2f}".format)})
result
```

Date	TR (USD/share)
2024-01-03	2.45
2024-01-04	3.37
2024-01-05	2.59
2024-01-08	4.42
2024-01-09	2.83
2024-01-10	2.48
2024-01-11	3.43
2024-01-12	1.55
2024-01-16	4.99
2024-01-17	3.33
2024-01-18	6.46
2024-01-19	3.32
2024-01-22	3.77
2024-01-23	1.92

The 14 values sum to 46.91 USD/share. Their mean is 3.3507. January 2 is excluded because its previous Close is outside this sample. No ATR14 is filled before January 23.

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Update ATR and interpret its size

After initialization, combine the previous ATR with today's TR. For January 24, TR is 2.04 and the previous ATR is 3.3507 USD/share.

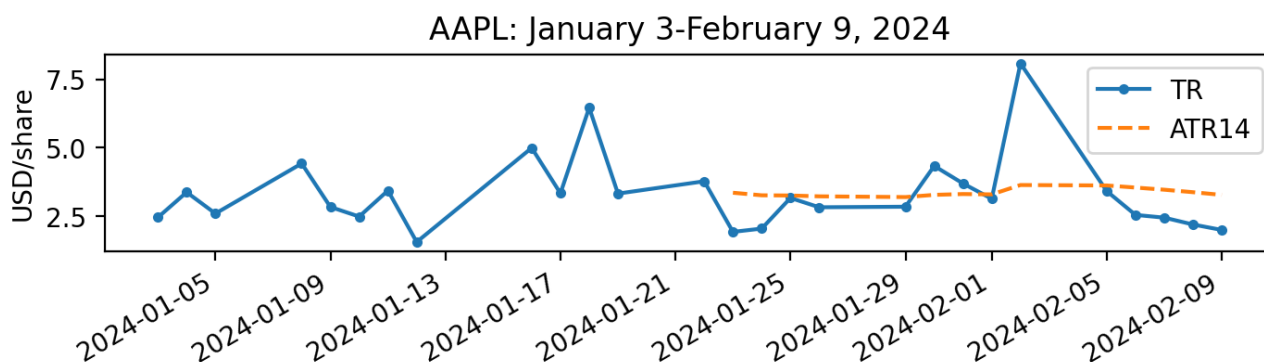
$$ATR_t = \frac{13 ATR_{t-1} + TR_t}{14}$$

$$ATR_{\text{Jan24}} \approx \frac{13(3.3507) + 2.0400}{14} \approx 3.2571 \text{ USD/share}$$

Before running: Will the January 24 TR pull ATR up or down?

```
def wilder_mean(values, n=14):
    assert len(values) >= n and np.isfinite(values).all()
    smoothed = pd.Series(np.nan, index=values.index)
    smoothed.iloc[n-1] = values.iloc[:n].mean()
    for i in range(n, len(values)):
        smoothed.iloc[i] = (
            (n-1)*smoothed.iloc[i-1] + values.iloc[i]) / n
    return smoothed

atr14 = wilder_mean(true_range)
fig, ax = plt.subplots(figsize=(7, 2.2))
dates = pd.to_datetime(true_range.loc["2024-02-09"].index)
ax.plot(dates, true_range.loc["2024-02-09"], "-.", label="TR")
ax.plot(dates, atr14.loc["2024-02-09"], "--", label="ATR14")
ax.set(title="AAPL: January 3-February 9, 2024", ylabel="USD/share")
ax.legend(); fig.autofmt_xdate(); plt.tight_layout()
```



ATR falls from 3.3507 to 3.2571 USD/share: 2.04 is below its prior average. A positive TR can lower ATR. Wilder smoothing retains earlier history; it is not a fresh rolling mean.

Common use: update a stop distance as ATR changes. Keep Close at 194.50 USD/share and the multiplier at 2. The distance falls from $2(3.3507) = 6.7014$ to $2(3.2571) = 6.5142$ USD/share, about USD 0.19 closer. These are hypothetical settings, not observed orders or guaranteed loss limits. [21]

Discuss

1. For January 24, explain why ATR falls even though TR is positive. Which two numbers decide the direction?
2. Would ATR 3.2571 alone tell you whether to buy or sell? Explain what it measures and what directional evidence is absent.

Sources: [19], [21]. Full titles and links are on the references page.

ISSUE 09

How is RSI calculated, and what does it tell us?

The Relative Strength Index (RSI) compares the size of recent upward and downward Close changes. Start by separating each change into a gain U and a loss magnitude D.

C is Close; delta C, U and D are USD/share. January 2 Close is 185.64. A loss of 1.39 becomes D=1.39, not a negative number.

$$\Delta C_t = C_t - C_{t-1}, \quad U_t = \max(\Delta C_t, 0), \quad D_t = \max(-\Delta C_t, 0)$$

$$\Delta C_{\text{Jan3}} = 184.25 - 185.64 = -1.39; \quad U_{\text{Jan3}} = 0, \quad D_{\text{Jan3}} = 1.39$$

Before running: On January 8, which of U and D is zero when Close rises from 181.18 to 185.56?

```
change = close.diff().iloc[1:]
gains, losses = change.clip(lower=0), (-change).clip(lower=0)
dates = change.index[:14]
result = pd.DataFrame({"Date": dates, "Close": close.loc[dates],
    "Change": change.loc[dates], "U": gains.loc[dates],
    "D": losses.loc[dates]}).round(2)
result
```

Date	Close	Change	U	D
2024-01-03	184.25	-1.39	0.0	1.39
2024-01-04	181.91	-2.34	0.0	2.34
2024-01-05	181.18	-0.73	0.0	0.73
2024-01-08	185.56	4.38	4.38	0.0
2024-01-09	185.14	-0.42	0.0	0.42
2024-01-10	186.19	1.05	1.05	0.0
2024-01-11	185.59	-0.6	0.0	0.6
2024-01-12	185.92	0.33	0.33	0.0
2024-01-16	183.63	-2.29	0.0	2.29
2024-01-17	182.68	-0.95	0.0	0.95
2024-01-18	188.63	5.95	5.95	0.0
2024-01-19	191.56	2.93	2.93	0.0
2024-01-22	193.89	2.33	2.33	0.0
2024-01-23	195.18	1.29	1.29	0.0

January 8 gain is 4.38, so U=4.38 and D=0. Each negative day contributes zero to U; each positive day contributes zero to D. Table prices and changes are USD/share.

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From daily gains and losses to RSI

U-bar and D-bar are Wilder-smoothed gains and losses, in USD/share. Initialize each with all 14 changes above, including zeros. Do not average only the up days or only the down days.

$$\bar{U}_{\text{Jan23}} = \frac{\sum_{i=1}^{14} U_i}{14}, \quad \bar{D}_{\text{Jan23}} = \frac{\sum_{i=1}^{14} D_i}{14}$$

$$\bar{U}_{\text{Jan23}} \approx 18.26/14 \approx 1.3043, \quad \bar{D}_{\text{Jan23}} \approx 8.72/14 \approx 0.6229$$

After initialization, apply the same update as ATR separately to gains and losses. On January 24, Close changes from 195.18 to 194.50: U=0, D=0.68.

$$\bar{U}_t = \frac{13\bar{U}_{t-1} + U_t}{14}, \quad \bar{D}_t = \frac{13\bar{D}_{t-1} + D_t}{14}$$

$$\bar{U}_{\text{Jan24}} \approx \frac{13(1.3043) + 0}{14} \approx 1.2111, \quad \bar{D}_{\text{Jan24}} \approx \frac{13(0.6229) + 0.68}{14} \approx 0.6269$$

RSI puts the smoothed upward part on a 0-100 scale. Dividing two dollar quantities removes the dollar unit. It is not a percentage investment return.

$$\text{RSI}_t = 100 \frac{\bar{U}_t}{\bar{U}_t + \bar{D}_t}$$

$$\text{RSI}_{\text{Jan24}} \approx 100 \frac{1.2111}{1.2111 + 0.6269} \approx 65.89$$

Date	U-bar	D-bar	RSI14
2024-01-23	1.3043	0.6229	67.68
2024-01-24	1.2111	0.6269	65.89
2024-01-25	1.1246	0.6057	64.99

January 25 Close is 194.17: U=0 and D=0.33 USD/share. Apply the same update to obtain the last row.

If D-bar=0 but U-bar>0, RSI=100; if U-bar=0 but D-bar>0, RSI=0. When both are zero, this lesson leaves RSI undefined. Initialization and flat-series conventions can differ across software.

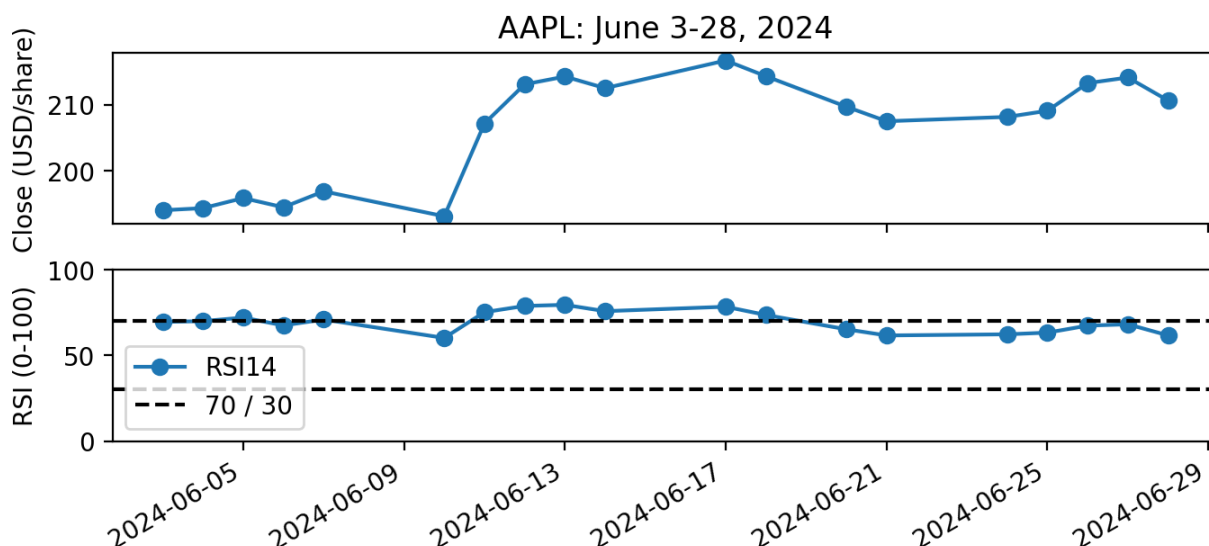
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Does RSI above 70 require a price fall?

70 and 30 are common overbought and oversold thresholds. They describe the indicator's level, not a guaranteed reversal. The code reuses Wilder smoothing from Issue 8.

Before running: Will Close fall on June 12 after June 11 RSI reaches 75.22?

```
up, down = wilder_mean(gains), wilder_mean(losses)
total = up + down
rsi14 = 100 * up / total.where(total > 0)
window = slice("2024-06-03", "2024-06-28")
dates = pd.to_datetime(close.loc[window].index)
fig, axes = plt.subplots(2, 1, figsize=(7, 3.3), sharex=True)
axes[0].plot(dates, close.loc[window], "o-")
axes[0].set(title="AAPL: June 3-28, 2024", ylabel="Close (USD/share)")
axes[1].plot(dates, rsi14.loc[window], "o-", label="RSI14")
axes[1].axhline(70, color="black", linestyle="--", label="70 / 30")
axes[1].axhline(30, color="black", linestyle="--")
axes[1].set(ylabel="RSI (0-100)", ylim=(0, 100))
axes[1].legend(); fig.autofmt_xdate(); plt.tight_layout()
```



June 11: Close 207.15 USD/share and RSI 75.22. Close rises to 213.07 on June 12 and 214.24 on June 13. RSI above 70 did not force an immediate fall in this case; this does not establish a profitable rule.

Common use: screen for overbought or oversold conditions before an entry or exit decision. An illustrative rule avoiding new purchases above RSI 70 would skip June 11 (75.22). The following rise shows its cost: avoiding a high RSI can also miss a continuing advance. [20]

Discuss

1. Under the update formula, if the next Close is unchanged, what happens to U-bar and D-bar? Must RSI change? Explain using their common multiplier.
2. Use June 11-13 to reject an immediate-fall claim. Why can these selected dates not establish how often the claim fails?

Sources: [20]. Full titles and links are on the references page.

ISSUE 10

How do volume changes compare with price returns?

A price return measures a price change. A volume change measures how the number of shares traded changed. The same ratio formula does not give them the same financial meaning.

V_t is daily Volume in shares; g_t is its decimal change from yesterday. C is Close in USD/share; r_t is its decimal price return. Previous denominators must be positive.

$$g_t = \frac{V_t}{V_{t-1}} - 1, \quad r_t = \frac{C_t}{C_{t-1}} - 1$$

$$g_{\text{Feb7}} = \frac{53,439,000}{43,490,800} - 1 \approx 22.87\%$$

Before running: Does a positive volume change imply a positive price return on every displayed date?

```
volume = aapl["Volume"]
daily_volume_change = volume / volume.shift(1) - 1
dates = close.loc["2024-02-01":"2024-02-09"].index
result = pd.DataFrame({"Date": dates,
    "Close (USD)": close.loc[dates].map("{:.2f}".format),
    "Volume (shares)": volume.loc[dates].map("{:, .0f}".format),
    "Price return": daily_return.loc[dates].map("{:.2%}".format),
    "Volume change": daily_volume_change.loc[dates].map("{:.2%}".format)})
result
```

Date	Close (USD)	Volume (shares)	Price return	Volume change
2024-02-01	186.86	64,885,400	1.33%	16.98%
2024-02-02	185.85	102,551,700	-0.54%	58.05%
2024-02-05	187.68	69,668,800	0.98%	-32.06%
2024-02-06	189.30	43,490,800	0.86%	-37.57%
2024-02-07	189.41	53,439,000	0.06%	22.87%
2024-02-08	188.32	40,962,000	-0.58%	-23.35%
2024-02-09	188.85	45,155,200	0.28%	10.24%

February 7 volume grows 22.87%, while its price return is 0.06%. These percentages describe different quantities. Neither tells us which traders initiated purchases or sales.

February 2 has volume change 58.05% and price return -0.54%. More trading occurred while Close fell. Volume growth is not money earned.

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More than yesterday, but less than usual?

Compare today's Volume with the mean of the preceding 20 observations, excluding today. The quotient is a multiple: 1 means equal to that mean, not a 100% return. i counts observations backward.

$$\frac{V_t}{\frac{1}{20} \sum_{i=1}^{20} V_{t-i}}$$

$$\frac{V_{\text{Feb7}}}{\text{prior 20 mean}} = \frac{53,439,000}{56,683,825} \approx 0.943$$

Before running: Can February 7 have a positive daily volume change and a multiple below 1?

```
prior20 = volume.shift(1).rolling(20, min_periods=20).mean()
assert (prior20.dropna() > 0).all()
volume_multiple = volume / prior20
dates = ["2024-02-06", "2024-02-07", "2024-02-08", "2024-02-09"]
result = pd.DataFrame({"Date": dates,
    "Volume (shares)": volume.loc[dates].map("{:, .0f}".format),
    "Prior mean": prior20.loc[dates].map("{:, .0f}".format),
    "Multiple": volume_multiple.loc[dates].map("{:.3f}x".format)})
result
```

Date	Volume (shares)	Prior mean	Multiple
2024-02-06	43,490,800	57,466,510	0.757x
2024-02-07	53,439,000	56,683,825	0.943x
2024-02-08	40,962,000	57,213,685	0.716x
2024-02-09	45,155,200	56,922,140	0.793x

February 7 traded 22.87% more shares than February 6, but only 0.943 times its prior-20 mean of 56,683,825 shares. Both statements hold: their comparison bases differ.

Common use: check whether a price rise comes with increased trading activity. [24] An illustrative rule requires a positive price return and Volume above its prior-20 mean. February 7 has +0.06% price return but only 0.943x Volume, so it fails. Comparing only with yesterday (+22.87%) would give a different answer.

This is a condition for considering a trade, not evidence that higher-volume signals earn more. Daily Volume does not identify who initiated the trades.

Discuss

1. Using February 6-7 Volume, reconstruct the 22.87% change. Why is it not a 22.87% investment gain?
2. For February 7, identify the comparison base for 22.87% and 0.943x. Why does shift(1) belong before the 20-observation mean?

Sources: [3], [15], [24]. Full titles and links are on the references page.

ISSUE 11

Are we counting the same calculation twice?

Suppose a rule counts one vote when EMA12 is below EMA26 and another when MACD is negative. Those are the same condition, so it has counted one calculation twice.

EMA12, EMA26 and MACD are in USD/share. Subtracting the longer EMA gives the exact equivalence below; it is not merely a high correlation.

$$E_t^{(12)} < E_t^{(26)} \iff E_t^{(12)} - E_t^{(26)} < 0 \iff \text{MACD}_t < 0$$

$$168.8635 < 170.4726 \iff -1.6092 < 0$$

Before running: Can the first two conditions disagree? Can the histogram condition differ?

```
dates = ["2024-04-24", "2024-04-25"]
result = pd.DataFrame({"Date": dates,
    "EMA12 < EMA26": [bool(ema12[d] < ema26[d]) for d in dates],
    "MACD < 0": [bool(macd[d] < 0) for d in dates],
    "Histogram > 0": [bool(histogram[d] > 0) for d in dates]})
result
```

Date	EMA12 < EMA26	MACD < 0	Histogram > 0
2024-04-24	True	True	False
2024-04-25	True	True	True

The first two conditions are True on both days. The histogram condition changes from False to True because it compares MACD with its signal line, not with zero.

Input	Calculation	Question
Close	EMA12 and EMA26	Which average is higher?
The two EMAs	MACD	What is their gap?
MACD and its EMA9	Histogram	Is MACD above its own average?

Derived features can answer different questions. Sharing the original Close data does not make every indicator a duplicate; an exact identity, as above, does.

Discuss

1. A rule requires both EMA12 < EMA26 and MACD < 0. Would removing either condition change its selected dates? Explain algebraically.
2. Why can the histogram condition differ on April 25 even though all calculations originate from Close? Does that make it an independent market observation?

Sources: [18], [23]. Full titles and links are on the references page.

ISSUE 12

Can we calculate today's SMA at 15:00?

Suppose a rule needs the three-price SMA ending January 4. At 15:00 New York time, today's regular-session Close does not yet exist: that session ends at 16:00.

Assume the earlier completed closes have arrived. C is Close in USD/share. These are the three inputs the rule needs, as seen at 15:00:

Close date	Value at 15:00	Can use it?
2024-01-02	185.64 USD	Yes
2024-01-03	184.25 USD	Yes
2024-01-04	Not yet known	No

$$SMA_{Jan4}^{(3)} = \frac{C_{Jan2} + C_{Jan3} + C_{Jan4}}{3}$$

The historical file later reports January 4 Close as 181.91. After that completed observation arrives, the calculation becomes:

$$SMA_{Jan4}^{(3)} = \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333 \text{ USD/share}$$

The correct numerical answer is not an available 15:00 input. A current intraday price could support a different rule, but it is not the final daily Close required here.

This is why a historical row's date is not enough: record when its required fields become available. Daily downloads do not identify the exact historical arrival time.

Discuss

1. Can replacing January 4 Close with January 3 Close preserve this rule? Write which three dates the modified average would contain.
2. After the final Close arrives, can the strategy claim an immediate fill at that same historical Close? Name the additional execution assumption that would need justification.

Sources: [1], [2]. Full titles and links are on the references page.

ISSUE 13

Which later return belongs to this feature row?

A feature uses information available now. A target is the later outcome to predict. Sharing a table row does not make both available now.

C_t is Close on the feature date and C_{t+1} the next observation's Close, both USD/share. y_t is the next-interval price-return target, in decimal units. For $t = \text{January 4}$, the target ends January 5.

$$y_t = \frac{C_{t+1}}{C_t} - 1$$

$$y_{\text{Jan4}} = \frac{181.18}{181.91} - 1 \approx -0.004013 = -0.4013\%$$

The 181.18 USD/share input is January 5 Close. It was not available with the January 4 features.

Before running: Why must the June 28 target remain missing in a sample ending June 28?

```
target = close.shift(-1) / close - 1
dates = ["2024-01-04", "2024-01-05", "2024-06-28"]
result = pd.DataFrame({"Feature date": dates,
    "Next-return target": ["Missing" if pd.isna(target[d])
        else f"{target[d]:.4%}" for d in dates]})
result
```

Feature date	Next-return target
2024-01-04	-0.4013%
2024-01-05	2.4175%
2024-06-28	Missing

January 4's target is -0.4013%. June 28 has no later Close in the sample, so its target is missing. A negative shift changes row placement, not when the outcome becomes known.

Use only targets observed by a training cutoff. Predicting this interval is not the same as earning its return; Week 4 adds execution and holdings.

Discuss

1. Would using target as a January 4 feature introduce future information? Name the price and date responsible.
2. At a June 28 training cutoff, may you use a subsequently downloaded July 1 Close to label June 28 as if that target were already known? Explain the timing.

Sources: [5], [6]. Full titles and links are on the references page.

ISSUE 14

Does a centered average use tomorrow's price?

A smoother historical line can use prices that had not occurred on its labeled date. Compare the inputs before treating that line as a trading feature.

C means Close in USD/share. For the value labeled January 4, trailing means today and two earlier observations; centered means yesterday, today and the next observation.

Date	Close (USD)	Trailing Jan 4	Centered Jan 4
2024-01-02	185.64	Included	Not used
2024-01-03	184.25	Included	Included
2024-01-04	181.91	Included	Included
2024-01-05	181.18	Not used	Included

$$SMA_t^{(3)} = \frac{C_{t-2} + C_{t-1} + C_t}{3}$$

$$SMA_{Jan4}^{(3)} = \frac{185.64 + 184.25 + 181.91}{3} \approx 183.9333$$

$$\text{Centered mean}_t = \frac{C_{t-1} + C_t + C_{t+1}}{3}$$

$$\text{Centered mean}_{Jan4} = \frac{184.25 + 181.91 + 181.18}{3} \approx 182.4467$$

Both answers are valid descriptions of the completed prices. Only the trailing answer can be calculated after January 4 Close arrives. The centered answer still needs January 5.

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Remove tomorrow and recompute today

Recompute January 4 using only data through that date. Keep the earlier prices and formula unchanged.

Before running: Which January 4 value survives when January 5 and all later rows are removed?

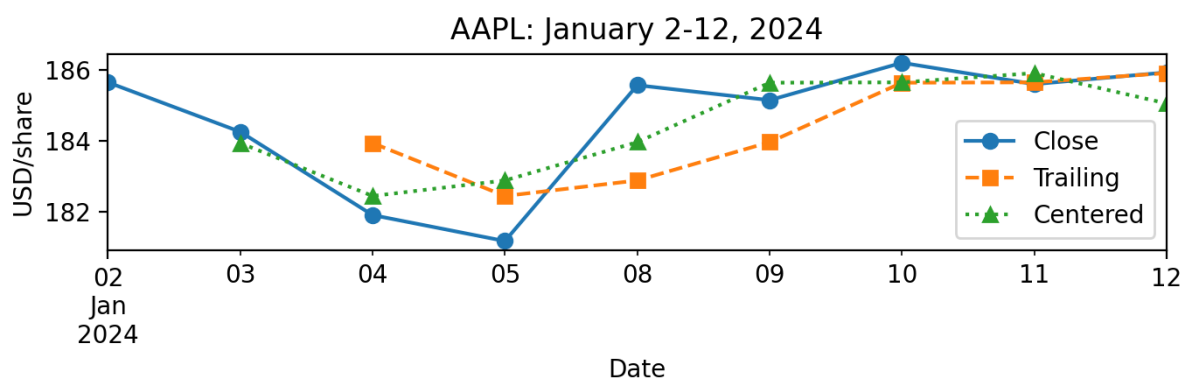
```
centered3 = close.rolling(3, min_periods=3, center=True).mean()
through_jan4 = close.loc["2024-01-04"]
prefix_centered = through_jan4.rolling(3, center=True).mean()
prefix_trailing = through_jan4.rolling(3).mean()
result = pd.DataFrame({"File ends": ["Jan 4", "Jun 28"],
    "Jan 4 trailing": [f"{prefix_trailing.iloc[-1]:.4f}",
        f"{sma3['2024-01-04']:.4f}"],
    "Jan 4 centered": ["Missing" if pd.isna(prefix_centered.iloc[-1])
        else f"{prefix_centered.iloc[-1]:.4f}",
        f"{centered3['2024-01-04']:.4f}"]})
result
```

File ends	Jan 4 trailing	Jan 4 centered
Jan 4	183.9333	Missing
Jun 28	183.9333	182.4467

Trailing stays 183.9333 USD/share; centered changes from 182.4467 to Missing. Its historical availability, not its arithmetic, is the problem.

Before running: Does being closer to January 4 Close make a line available earlier?

```
lines = pd.DataFrame({"Close": close, "Trailing": sma3,
    "Centered": centered3}).iloc[:9]
lines.index = pd.to_datetime(lines.index)
ax = lines.plot(style=["o-", "s--", "^:"], figsize=(7, 2.4))
ax.set(title="AAPL: January 2-12, 2024", ylabel="USD/share")
plt.tight_layout()
```



Centered 182.4467 is closer to Close 181.91 than trailing 183.9333. It uses a later price, so this closeness is not forecasting evidence.

Discuss

1. Which later price is needed for the centered January 4 value? When does it become available?
2. Why must trailing SMA stay 183.9333 when later rows are removed?

Sources: [3], [6]. Full titles and links are on the references page.

ISSUE 15

Which data may set the scaling parameters?

Standardize April 5 momentum using a mean and standard deviation fitted through April 4. Including later data would change what was available on April 5.

$m_t^{(3)}$ is three-interval decimal momentum. μ_{fit} is its fitted mean; σ_{fit} is its positive, finite sample standard deviation (ddof=1). z_t is the number of fitted standard deviations from that mean.

$$z_t = \frac{m_t^{(3)} - \mu_{\text{fit}}}{\sigma_{\text{fit}}}$$

$$z_{\text{Apr5}} = \frac{0.0043829 - (-0.0038024)}{0.0227433} \approx 0.3599$$

The mean and standard deviation use 62 momentum observations through April 4. Freeze them before transforming the April 5 input 0.0043829.

Before running: Which column uses observations unavailable for an April 5 decision? Are the two columns transforming the same input?

```
fit = momentum3.loc["2024-04-04"].dropna()
mu, scale = fit.mean(), fit.std(ddof=1)
assert np.isfinite(scale) and scale > 0
apr5 = momentum3.loc["2024-04-05"]
z_earlier = (apr5 - mu) / scale
z_full = (apr5 - momentum3.mean()) / momentum3.std(ddof=1)
result = pd.DataFrame({"Date": ["2024-04-05"],
    "Fit through Apr 4": [f"{z_earlier:.4f}"],
    "Fit through Jun 28": [f"{z_full:.4f}"]})
result
```

Date	Fit through Apr 4	Fit through Jun 28
2024-04-05	0.3599	0.0150

The same April 5 momentum becomes 0.3599 under the earlier fit and 0.0150 under the full-file fit. These are standardized units, not percent returns. Only the fitting window changed.

These parameters are usable after the April 4 data arrive and the fit finishes, not at every earlier date in the training sample. z is dimensionless, not a return.

Discuss

1. Why is the full-file result invalid for an April 5 real-time feature even if it is numerically close to the earlier-fit result?
2. Can the April 4 parameters be assigned to a January 5 historical decision as if already known then? Distinguish offline training from a contemporaneous decision.

Sources: [6], [14]. Full titles and links are on the references page.

ISSUE 16

How does a feature become a signal?

A signal is a specified condition, not a trade. A common golden cross is SMA50 crossing above SMA200; a death cross is the reverse. Being above is a state; crossing above is an event.

Both SMAs use Close in USD/share. Check yesterday and today. For a golden cross, require both inequalities; reverse them for a death cross.

$$SMA_{t-1}^{(50)} \leq SMA_{t-1}^{(200)}, \quad SMA_t^{(50)} > SMA_t^{(200)}$$

A 200-price SMA needs earlier history. Extend AAPL to January 2023 for Issues 16-17; the overlapping 2024 closes remain unchanged.

Before running: Which dates change the ordering of SMA50 and SMA200?

```
long_history = yf.Ticker("AAPL").history(start="2023-01-01",
    end="2024-07-01", interval="1d", auto_adjust=False, actions=True)
long_history.index = long_history.index.strftime("%Y-%m-%d")
long_close = long_history["Close"]
assert np.allclose(long_close.loc[close.index], close, rtol=0, atol=1e-10)
sma50, sma200 = long_close.rolling(50).mean(), long_close.rolling(200).mean()
golden = (sma50 > sma200) & (sma50.shift(1) <= sma200.shift(1))
death = (sma50 < sma200) & (sma50.shift(1) >= sma200.shift(1))
dates = ["2024-03-12", "2024-03-13", "2024-06-12", "2024-06-13"]
result = pd.DataFrame({"Date": dates,
    "SMA50": sma50.loc[dates].map("{:.4f}".format),
    "SMA200": sma200.loc[dates].map("{:.4f}".format),
    "Cross": ["Golden" if golden[d] else "Death" if death[d]
        else "None" for d in dates]})
result
```

Date	SMA50	SMA200	Cross
2024-03-12	184.1992	183.8796	None
2024-03-13	183.7712	183.8703	Death
2024-06-12	181.5936	182.1121	None
2024-06-13	182.4854	182.2824	Golden

June 12: 181.5936 is below 182.1121. June 13: 182.4854 is above 182.2824 USD/share, a golden cross. March 13 reverses the ordering, a death cross. These names do not guarantee later returns.

Common use: consider entry after a golden cross, or exit after a death cross. This would flag June 13 for entry and March 13 for exit, after the closes arrive. [24] It is not a completed trade; missing SMA200 history cannot establish a cross.

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Combine explicit indicator conditions

Illustrative rule: set $s_t=1$ when MACD is above its signal line and RSI14 is below 70; otherwise set 0. Keep Missing when either input is unavailable. The rule is not an investment recommendation.

$$s_t = \mathbf{1}\{\text{MACD}_t > S_t \text{ and } \text{RSI}_t^{(14)} < 70\}$$

The indicator function is 1 when both conditions hold, 0 otherwise. S is the MACD signal line from Issue 4, not the trading signal s.

Before running: Will a positive histogram alone make the signal 1 on June 11?

```
signal = pd.Series(np.nan, index=close.index)
valid = histogram.notna() & rsi14.reindex(close.index).notna()
signal.loc[valid] = ((histogram.loc[valid] > 0)
    & (rsi14.loc[valid] < 70)).astype(int)
dates = ["2024-05-06", "2024-06-07", "2024-06-10", "2024-06-11"]
result = pd.DataFrame({"Date": dates,
    "Histogram (USD)": histogram.loc[dates].map("{:.4f}".format),
    "RSI14": rsi14.loc[dates].map("{:.2f}".format),
    "Signal": signal.loc[dates].astype(int)})
result
```

Date	Histogram (USD)	RSI14	Signal
2024-05-06	1.6806	63.79	1
2024-06-07	0.0507	70.92	0
2024-06-10	-0.1925	60.18	0
2024-06-11	0.5214	75.22	0

May 6 satisfies both conditions ($1.6806 > 0$ and $63.79 < 70$), so $s=1$. June 11 has a positive histogram but RSI 75.22 fails the second condition, so $s=0$. These are rule outputs, not forecasts that prices must rise or fall.

SMA, MACD and RSI can supply directional conditions. ATR supplies movement size, not direction by itself. A signal still needs an execution time, price and position rule before it becomes a backtest; those follow in Week 4.

Discuss

1. Using June 12-13 SMA values, show why June 13 is a crossing event rather than merely a day with SMA50 above SMA200.
2. For June 11, compare the AND rule with an OR rule using the table. Would the signal change? Explain why missing RSI must not be filled with zero.

Sources: [1], [18], [20], [24]. Full titles and links are on the references page.

ISSUE 17

Why can earnings growth and the share price disagree?

A financial report summarizes an earlier business period. A daily share price is a different quantity observed on a different date. Earnings growth therefore need not equal the next share-price return.

Apple announced these results on February 1, 2024. Net sales are USD million; diluted earnings per share (EPS) are USD/share. The two fiscal quarters contain different numbers of weeks.

Quarter ended	Weeks	Sales (USD m)	EPS (USD)
2022-12-31	14	117,154	1.88
2023-12-30	13	119,575	2.18

EPS growth compares this reported quarter with the previous fiscal year's corresponding quarter; it is not the return on holding one share.

$$\text{EPS growth} = \frac{\text{EPS}_{\text{new}}}{\text{EPS}_{\text{old}}} - 1$$

$$\text{EPS growth} = \frac{2.18}{1.88} - 1 \approx 15.96\%$$

Date	What happened
2023-12-30	Reported business period ended
2024-02-01	Company announced results
2024-02-01 18:03:38	SEC accepted the 10-Q
2024-02-02	SEC filing date

December 30 identifies the business period, not when these results became public. The company announcement and SEC filing are separate sources; SEC acceptance is not automatically the first disclosure of every number.

Use the actual publication record for the field you need. These records do not establish each investor's receipt time or an executable trade price.

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Place the announcement on the price chart

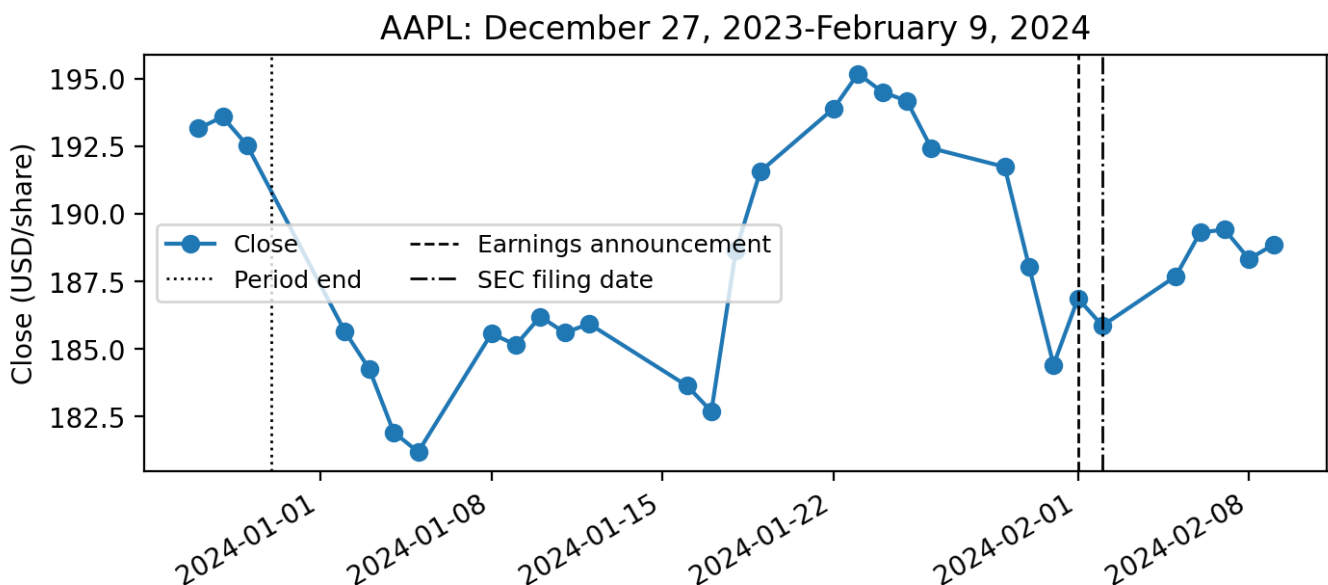
Use the extended AAPL history from Issue 16. The chart aligns business-period end, announcement date and filing date with daily Close; it does not attribute the price change to one cause.

Before running: Must a 15.96% year-over-year EPS increase imply a positive February 1-2 Close return?

```

window = long_close.loc["2023-12-27":"2024-02-09"]
fig, ax = plt.subplots(figsize=(7, 3.3))
ax.plot(pd.to_datetime(window.index), window, "o-", label="Close")
for date, label, style in [
    ("2023-12-30", "Period end", ":"),
    ("2024-02-01", "Earnings announcement", "--"),
    ("2024-02-02", "SEC filing date", "-.")]:
    ax.axvline(pd.Timestamp(date), color="black", linestyle=style,
               linewidth=1, label=label)
ax.set(title="AAPL: December 27, 2023-February 9, 2024",
       ylabel="Close (USD/share)")
ax.legend(fontsize=9, ncol=2)
fig.autofmt_xdate(); plt.tight_layout()

```



February 1 Close is 186.86; February 2 is 185.85 USD/share. The price return is $(185.85/186.86)-1$, approximately -0.54%, despite the positive reported EPS growth.

The comparisons have different periods and meanings. This one event shows that their signs can differ; it does not identify expectations, isolate a news effect or establish a trading rule. A daily chart also cannot resolve the announcement's intraday timing.

Discuss

1. Could the December 30 period-end label justify using EPS 2.18 in a January 2 strategy? Identify the later publication evidence.
2. Compare +15.96% EPS growth with -0.54% Close return. State each numerator, denominator and period; explain why the figures are not contradictory.

Sources: [7], [22], [25]. Full titles and links are on the references page.

ISSUE 18

Which GDP estimate was known then?

GDP measures economic activity, not company earnings. Q4 2023 ended in December, but its first GDP growth estimate appeared in January and was revised in February.

Both BEA figures are seasonally adjusted, annualized real GDP growth from the preceding quarter. They are not stock returns. Both releases were scheduled for 08:30 New York time on their dates.

Release date	Quarter	Growth (%)
2024-01-25	2023 Q4	3.3
2024-02-28	2023 Q4	3.2

The estimate changes by $3.2 - 3.3 = -0.1$ percentage point. That is a revision to the same quarter, not growth from January to February. Newer estimates exist; retain these historical versions for the dated comparison.

Before running: Using only releases earlier than each hypothetical decision date, should February 1 and March 1 receive the same estimate?

```
# Values transcribed from the two dated BEA releases.
releases = pd.DataFrame({"Release": pd.to_datetime(
    ["2024-01-25", "2024-02-28"]), "GDP (%)": [3.3, 3.2]})
decisions = pd.DataFrame({"Decision": pd.to_datetime(
    ["2024-02-01", "2024-03-01"])})
matched = pd.merge_asof(decisions, releases,
    left_on="Decision", right_on="Release", direction="backward")
result = matched.copy()
for field in ["Decision", "Release"]:
    result[field] = result[field].dt.strftime("%Y-%m-%d")
result
```

Decision	Release	GDP (%)
2024-02-01	2024-01-25	3.3
2024-03-01	2024-02-28	3.2

February 1 receives 3.3%; March 1 receives 3.2%. Each decision date is later than the relevant release day. The code picks the latest earlier release, not the newest value in today's database.

Publication delay and revision differ from the smoothing delay in an SMA. No moving average is being calculated here.

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Compare release dates with daily prices

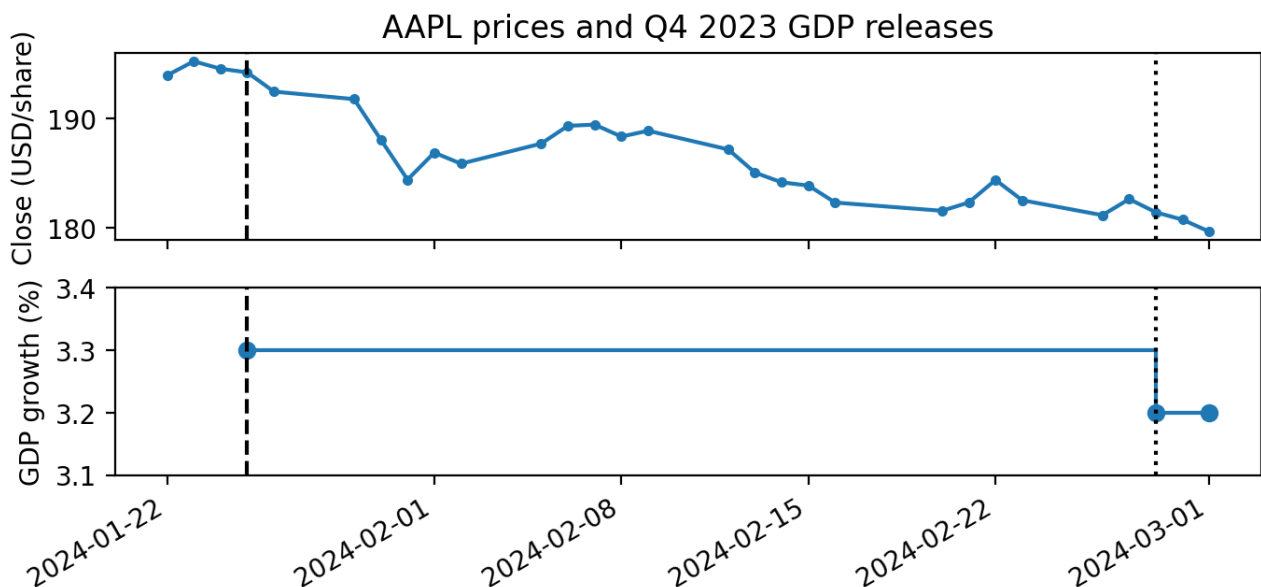
The lower panel shows the Q4 estimate available after each release; the upper panel shows AAPL Close. Separate axes preserve the different units. There is no Q4 estimate plotted before January 25.

Before running: Will the GDP line change every day just because the price line does?

```

window = close.loc["2024-01-22":"2024-03-01"]
fig, axes = plt.subplots(2, 1, figsize=(7, 3.4), sharex=True)
axes[0].plot(pd.to_datetime(window.index), window, "o-", markersize=3)
axes[0].set(title="AAPL prices and Q4 2023 GDP releases",
            ylabel="Close (USD/share)")
release_dates = pd.to_datetime(["2024-01-25", "2024-02-28", "2024-03-01"])
axes[1].step(release_dates, [3.3, 3.2, 3.2], where="post", marker="o")
axes[1].set(ylabel="GDP growth (%)", ylim=(3.1, 3.4))
for ax in axes:
    ax.axvline(release_dates[0], color="black", linestyle="--")
    ax.axvline(release_dates[1], color="black", linestyle=":")
fig.autofmt_xdate(); plt.tight_layout()

```



The GDP estimate stays 3.3% until the February 28 revision. AAPL Close is 194.17 on January 25 and 181.42 USD/share on February 28. These series describe different objects and frequencies; their joint movement is not evidence that the revision caused the price decline.

The step marks publication dates, not a gradual deterioration in GDP. For an earlier decision, preserve the version available then; do not copy 3.2% backward into January.

Discuss

1. On February 1, which Q4 GDP value was available? Explain why using 3.2% would introduce information from a later date.
2. Why can the price line move while the GDP line is flat? Distinguish an unchanged published estimate from an unchanged economy or company value.

Sources: [8], [9], [16]. Full titles and links are on the references page.

REFERENCES & REPRODUCTION

Sources and calculation notes

- [1] [yfinance historical-data interface](#)
- [2] [Nasdaq market hours](#)
- [3] [pandas rolling windows](#)
- [4] [pandas exponentially weighted calculations](#)
- [5] [pandas shift](#)
- [6] [scikit-learn: data leakage](#)
- [7] [Apple Form 10-Q filing record](#)
- [8] [BEA: Q4 2023 GDP advance estimate](#)
- [9] [BEA: Q4 2023 GDP second estimate](#)
- [10] [Fidelity: simple moving average](#)
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- [12] [Lo, Mamaysky and Wang: technical analysis research](#)
- [13] [Fidelity: rate of change](#)
- [14] [NIST: standard deviation](#)
- [15] [Nasdaq glossary: volume](#)
- [16] [pandas: match earlier observations](#)
- [17] [Yahoo Finance: AAPL history and price-field notes](#)
- [18] [Fidelity: MACD](#)
- [19] [StockCharts: ATR calculation and initialization](#)
- [20] [StockCharts: RSI calculation and interpretation](#)
- [21] [Fidelity: True Range and ATR](#)
- [22] [Apple: February 1, 2024 earnings announcement](#)
- [23] [TradingView: MACD and histogram](#)
- [24] [Fidelity: technical analysis and common indicator uses](#)
- [25] [Apple: FY2024 Q1 financial statements, page 1](#)
- [26] [Fidelity: historical return variability and investment risk](#)
- [27] [Investor.gov: stop orders and execution-price limits](#)

How the sources are used

Market observations come from Yahoo Finance through yfinance. Only selected inputs and derived results are shown; the full download is not redistributed. SEC and BEA support dated document and release facts. Software documentation supports calculation behavior, not prediction or profitability.

SMA, EMA and rate-of-change sources identify established indicators. Our short-window examples are not recommendations or replications of the Lo, Mamaysky and Wang research design. Figures describe observed data only.

Executed 2026-09-21 UTC: Python 3.12.9, numpy 2.2.6, pandas 2.2.3, matplotlib 3.10.8, yfinance 1.2.0. Run all displayed code in order in a fresh Python notebook. Install the named packages if absent. Data access requires internet and provider availability.

Historical responses may change. Receipt times and point-in-time market-data versions were not independently verified. The checked calculations do not fill those evidentiary gaps. This lesson does not assign weights, fills, costs or trading performance.